SUBJECT NAME: COMPUTER ORGANIZATION AND ARCHITECTURE

SUBJECT CODE: SECR1033 / SCSR1033

SEMESTER: 2 – 2025/2026

LAB TITLE: Programming 1: Assembly Language Fundamentals

INSTRUCTION: Student is required to have instructor's signature before proceed from each lab work.

STUDENT INFO :

Group member 1:	Name	: CHENG ZHI MIN
	Metric No	: A25CS0050
	Section	: 03
	Email	: chengzhimin@graduate.utm.my
Group member 2:	Name	: NG XUAN YEE
	Metric No	: A25CS0291
	Section	: 03
	Email	: ngxuanyee@graduate.utm.my

SUBMITTED DATE:

COMMENTS:

Lab # 1

Simple program to familiarize with Program Code, Rebuild & Start Without Debugging

Execute the programs below:
PART 1:

i. **Part A**: Adding and Subtracting Integers

```
TITLE Add and Subtract (AddSub.asm)

; This program adds and subtracts 32-bit integers
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc

.data
TOTAL dword 0 ; a variable named TOTAL (declared as DWORD)

.code

main PROC

    mov eax, 123400h ; Set EAX with the value of 123400h
    add eax, 567800h ; Add the content of EAX with 567800h
    sub eax, 77700h ; Subtract content of EAX with 77700h
    mov TOTAL, eax  ; Store content of EAX to TOTAL

    call DumpRegs

    exit
main ENDP

END main
```

Experimental Results:
a) Rebuild & Start Without Debugging

Completed *Fully* *Partially* *: Checked by:*_____

Screenshot Result:

_____

Microsoft Visual Studio Debu

```
  EAX=00613500  EBX=00A5B000  ECX=008910FF  EDX=008910FF
  ESI=008910FF  EDI=008910FF  EBP=00CFF9C0  ESP=00CFF9B4
  EIP=00893679  EFL=00000206  CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1


C:\Users\USER\source\repos\lab1a\Debug\lab1a.exe (process 2636) exited with code 0 (0x0).
Press any key to close this window . . .
```

ii. **Part B**: Adding Variables

```
TITLE Add and Subtract, Version 2(AddSub2.asm)

; This program adds and subtracts 32-bit unsigned
; integers and stores the sum in a variable.
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc
.data
val1 DWORD 10000h
val2 DWORD 40000h
val3 DWORD 20000h
finalVal DWORD ?
.code

main PROC
mov eax,val1      ; start with 10000h
add eax,val2      ; add 40000h
sub eax,val3      ; subtract 20000h
mov finalVal,eax ; store the result (30000h)

call DumpRegs     ; display the registers

exit

main ENDP
END main
```

Experimental Results:

a) Rebuild & Start Without Debugging

Completed *Fully* *Partially* *: Checked by:*_____

Screenshot Result:

Microsoft Visual Studio Debu

```
  EAX=00030000  EBX=010A9000  ECX=00EB10FF  EDX=00EB10FF
  ESI=00EB10FF  EDI=00EB10FF  EBP=0133FE80  ESP=0133FE74
  EIP=00EB367B  EFL=00000206  CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1


C:\Users\USER\source\repos\lab1b\Debug\lab1b.exe (process 10036) exited with code 0 (0x0).
Press any key to close this window . . .
```

iii. **Part C**: Add and Subtract 8 and 16-Bit Version

```
TITLE Add and Subtract, Version 3
; This program adds and subtracts 8 and 16 bit
; unsigned integers and stores the sum in a variable.
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc
.data
valw1 WORD 1000h
valw2 WORD 4000h
valw3 WORD 2000h
finalValw WORD ?

valb1 BYTE 10h
valb2 BYTE 40h
valb3 BYTE 20h
finalValb BYTE ?

.code
main PROC
mov ax,valw1; start with 10000h
add ax,valw2       ; add 40000h
sub ax,valw3       ; subtract 20000h
mov finalValw,ax  ; store the result (30000h)
call DumpRegs      ; display the registers

mov ah,valb1; start with 10000h
add ah,valb2       ; add 40000h
sub ah,valb3       ; subtract 20000h
mov finalValb,ah  ; store the result (30000h)
call DumpRegs      ; display the registers

exit
main ENDP
END main
```

Experimental Results:

a) Rebuild & Start Without Debugging

Completed *Fully* *Partially* *: Checked by:* _____

Screenshot Result:

Microsoft Visual Studio Debu

```
  EAX=00D63000  EBX=00A40000  ECX=001F10FF  EDX=001F10FF
  ESI=001F10FF  EDI=001F10FF  EBP=00D6F960  ESP=00D6F954
  EIP=001F367F  EFL=00000206  CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1


  EAX=00D63000  EBX=00A40000  ECX=001F10FF  EDX=001F10FF
  ESI=001F10FF  EDI=001F10FF  EBP=00D6F960  ESP=00D6F954
  EIP=001F369C  EFL=00000206  CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1


C:\Users\USER\source\repos\lab1c\Debug\lab1c.exe (process 8184) exited with code 0 (0x0).
Press any key to close this window . . .
```

PART 2 Video

part b.mp4

part c.mp4

What is Byte Code (or Machine Code) for the following assembly instructions:

```
mov ax,valw1  ; start with 10000h
add ax,valw2        ; add 40000h
sub ax,valw3        ; subtract 20000h
mov finalValw,ax    ; store the result (30000h)
```

Byte Code:

```
--- C:\Users\USER\source\repos\lab1c\main.asm ---------------------------
mov ax, valw1; start with 10000h
002E3660 66 A1 00 60 2E 00    mov         ax,word ptr [valw1 (02E6000h)]
add ax, valw2; add 40000h
002E3666 66 03 05 02 60 2E 00 add         ax,word ptr [valw2 (02E6002h)]
sub ax, valw3; subtract 20000h
002E366D 66 2B 05 04 60 2E 00 sub         ax,word ptr [valw3 (02E6004h)]
mov finalValw, ax; store the result(30000h)
002E3674 66 A3 06 60 2E 00    mov         word ptr [finalValw (02E6006h)],ax
```